

EE2080: Random Processes, Exam 1

Max. Marks: 30. **Time:** 2 hours.

Instructions

- Please **write your serial number and roll number** prominently in the first page of the answer sheet. Please see the attendance sheet to know your serial number.
- No breaks during the exam. No laptops, calculators, mobile devices, smart watches etc. are allowed.
- Each step in your answer must be supported by clear mathematical reasoning. For questions requiring you to provide proofs, you must include a precise and rigorous proof.
- **Unless otherwise stated, any result proved in the lectures can be used by stating it clearly, no need to prove them in your answer. Any other result must be proved before using them in your answer.**
- Only the best possible answer (for example, the most-simplified form) to a question will be awarded the full points.

Q1: (15 marks) Consider the gambler's ruin problem with draws. Alice starts with \$ a , and Bob starts with \$ $(M - a)$. At each time step,

- Alice wins \$1 from Bob with probability p ,
- Alice loses \$1 to Bob with probability q ,
- No money is exchanged with probability s ,

where $p + q + s = 1$.

Assume Alice and Bob are equally matched, so that $p = q$, $s = 1 - 2p$, where $0 < p < \frac{1}{2}$.

The game continues until either Alice is ruined or Bob is ruined. Let r_i denote the probability that Alice is eventually ruined given that she currently has \$ i .

- (4 marks)** Show that $pr_{i+1} - (1 - s)r_i + pr_{i-1} = 0$, and state the appropriate boundary conditions. Then, solve the linear difference equation to obtain an expression for r_i . (Hint: assume $r_i = \lambda^i$, solve the characteristic equation, if the roots are x and y , then the solution is $r_i = Ax^i + By^i$ and find A and B using boundary conditions.)
- (2 marks)** Let X_n denote Alice's fortune after n rounds, with $X_0 = a$. Consider only those rounds in which money is actually exchanged. Let Z_k denote Alice's fortune immediately after the k th such round. Show that $\{Z_k\}$ is a simple symmetric random walk on $\{0, 1, \dots, M\}$, absorbed at 0 and M .
- (3 marks)** Let N denote the number of money-exchange rounds required for the game to end. For the simple symmetric random walk $\{Z_k\}$, you may use without proof the identity

$$\mathbb{E}[N] = \mathbb{E}[Z_N^2] - a^2.$$

Using this identity and your result from part (a), derive $\mathbb{E}[N]$.

- (3 marks)** Let G_k denote the number of original rounds from the $(k - 1)$ st money exchange up to and including the k th money exchange. Determine the distribution of G_k and compute $E[G_k]$. Are $G_1, G_2, \dots$ identically distributed and independent?
- (3 marks)** Let T denote the total number of rounds until the game ends. Express T in terms of N and $G_1, G_2, \dots$ and then using it, find $E[T]$, the expected duration of the original game with draws. (Justify every step.)

Q2: (2 marks) Let $\{S_n\}_{n \in \mathbb{N}}$ be a Poisson process with rate λ . Here, S_n is the n^{th} arrival time. Let $\{N_t\}_{t \geq 0}$ be the associated counting process. Show that given $N_t = 1$, $S_1 \sim \text{unif}(0, t]$, i.e., prove that $P(S_1 \leq x | N_t = 1) = \frac{x}{t}$, for $0 \leq x \leq t$.

Q3: (3 marks) Let $X_1, X_2, \dots$ be i.i.d. random variables, each uniformly distributed on $[0, 1]$. For each positive integer k , define $N_k = \min \{n \geq 1 : \sum_{i=1}^n X_i > k\}$. Evaluate $\lim_{k \rightarrow \infty} \frac{\mathbb{E}[N_k]}{k}$.

Q4: (5 marks) Let $\{X[n]\}_{n \in \mathbb{Z}}$ be a wide-sense stationary (WSS) random process with mean μ , autocorrelation function $R_X[k]$, and autocovariance function $C_X[k]$.

(a) **(3 marks)** Define the time-average estimate of the mean based on N observations as

$$\bar{X}_N = \frac{1}{N} \sum_{n=0}^{N-1} X[n].$$

Show that $E[\bar{X}_N] = \mu$, and prove that $X[n]$ is ergodic in mean if and only if $\lim_{N \rightarrow \infty} \text{Var}(\bar{X}_N) = 0$. Further, express $\text{Var}(\bar{X}_N)$ in terms of $C_X[k]$ and hence obtain a condition on the autocovariance function for mean ergodicity.

(b) **(1 mark)** For a fixed lag k , define the time-average estimate of the autocorrelation as

$$\hat{R}_N[k] = \frac{1}{N} \sum_{n=0}^{N-1} X[n]X[n+k].$$

Is WSS alone sufficient to conclude that $\hat{R}_N[k] \xrightarrow{\text{m.s.}} R_X[k]$? Briefly justify your answer by identifying what additional statistical information would be required. (Hint: Expand $\text{Var}(\hat{R}_N[k])$.)

(c) **(1 mark)** How does the analysis in Part (b) simplify if $X[n]$ is a WSS Gaussian process?

Q5: (5 marks) Let $X_1, X_2, \dots$ be IID exponential random variables with parameter $\lambda > 0$, and define the arrival times $S_n = \sum_{i=1}^n X_i$. Let the associated counting process be $N(t) = \max\{n : S_n \leq t\}$, $t \geq 0$.

(a) **(2 marks)** For a fixed $t > 0$, let

$$R_t = S_{N(t)+1} - t$$

denote the waiting time from t until the first arrival after t . Show that $R_t \sim \text{Exp}(\lambda)$.

(b) **(1 mark)** Explain why part (a) implies that the arrival process observed after time t has the same distribution as a fresh copy of the process starting at time 0.

(c) **(2 marks)** Show that the process $\{N(t+u) - N(t) : u \geq 0\}$ has the same distribution as $\{N(u) : u \geq 0\}$. Hence, prove that $\{N(t)\}$ has **stationary increments**; i.e., for all $t, h \geq 0$, $N(t+h) - N(t) \stackrel{d}{=} N(h)$.